

AI Safety & Governance Layer

Team Name : PERCEPTRON

College Name : Panimalar Engineering College

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PROBLEM STATEMENT DETAILING

- Enterprises deploying AI/agent systems risk leaking PII and sensitive data through prompts, tool calls, and responses, with no visibility into what was exposed. Goal: give teams a real-time layer that catches and contains this exposure.
- Sub-tasks: (1) detect PII/sensitive entities in real time across prompts and responses, (2) mask or redact before data leaves the boundary, (3) generate tamper-evident audit logs. Challenge: do this with low latency inside live agent pipelines.
- Success measures: PII detection accuracy (precision/recall), masking latency added per request, % of sensitive events captured in audit logs, false-positive rate on legitimate content.

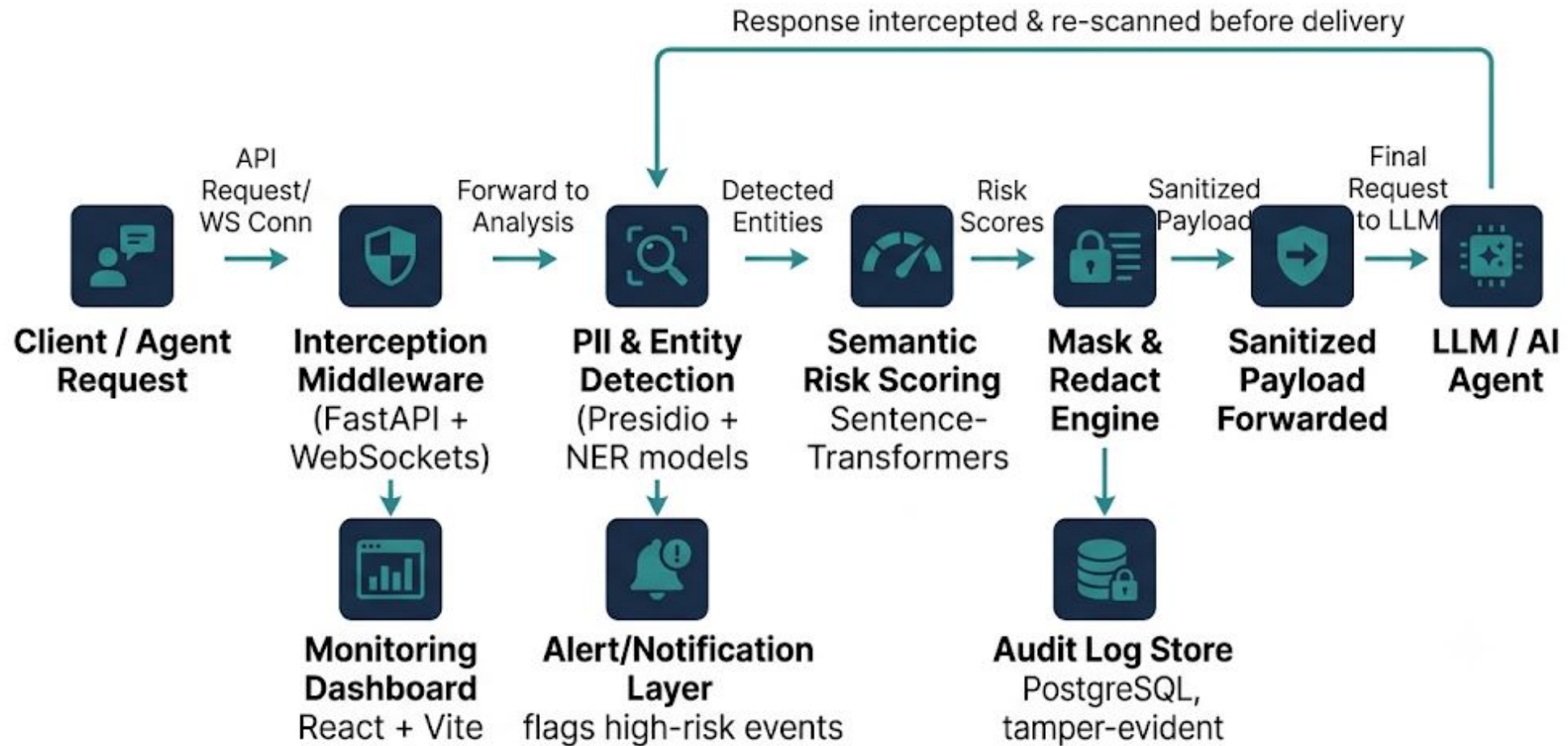
SOLUTION

- The firewall layer intercepts every prompt/response, flags and masks PII before exposure, and logs every decision — directly reducing leaked-PII incidents and giving full audit traceability.
- Technology stack:
- FastAPI + WebSockets (real-time interception), Microsoft Presidio (PII detection/masking), Sentence-Transformers (semantic risk scoring), React/Vite (dashboard), PostgreSQL (audit log store)
- High-level design:
Client request → Interception middleware → PII/entity detection (Presidio + NER) → Mask/redact → Audit log write → Forward sanitized payload to LLM/agent → Response scanned on the way back
- Dependencies: the LLM/agent API being protected, Presidio's NER models, and a persistent store for audit logs.
- Limitations: added latency per request, coverage limited to entity types the NER model recognizes, no protection against out-of-band data exfiltration.

SOURCE CODE

SabarishR08/ai-safety-governance-layer: Real-time PII detection, masking, and audit-log firewall for AI/agent systems — PEC Techathon 4.0

Architecture Diagram



User Interface

Mentor connect details (Kindly add the details of the connect had with the mentors)

Date of connect	Time	Details on the development	Agenda of the meet
DD-MM-YY	HH:MM		Meeting agenda on the topic

Thank you

Additional Information
<Any relevant attachment>

